

Olympiad Test Paper — Science (Class 7) — Paper 3

Chapter 2: Nutrition in Animals

Paper 3 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	2 — Nutrition in Animals
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Traps target the egestion/excretion and absorption/assimilation confusions, the bile-is-an-enzyme and HCl-digests-fat errors, the rumen-vs-abomasum and cud-vs-chyme mix-ups, surface-area reasoning about villi, and the single-cell-does-all-five-steps idea for Amoeba. Do not write on this paper — mark answers on your answer sheet.

1. A child eats a banana. Glucose from it appears in the child's blood about an hour later, and a week afterwards the child has grown slightly taller using material partly from that glucose. The blood-appearance event and the height-gain event are, in order, examples of which two steps of nutrition?

- (A) Digestion then absorption (B) Absorption then egestion
(C) Assimilation then absorption (D) Absorption then assimilation

2. A doctor describes a patient: 'Food is taken in and broken down normally, the lining of the gut is healthy, but the patient's cells seem unable to use the glucose that reaches them, so the patient is weak despite eating.' Which single step of nutrition is failing, and which is clearly working?

- (A) Absorption is failing; digestion is working
(B) Assimilation is failing; absorption is working
(C) Digestion is failing; ingestion is working
(D) Egestion is failing; assimilation is working

3. Two waste-removal processes are compared. Process X removes leftover fibre and dead gut cells from the food canal; process Y removes substances such as carbon dioxide made when cells release energy. Which statement is correct?

- (A) Both X and Y are egestion, differing only in name
- (B) X is excretion of cell waste; Y is egestion of undigested food
- (C) X is egestion and removes undigested matter; Y is excretion and removes wastes made inside cells
- (D) X removes cell waste; Y removes undigested food

4. A snake swallows a whole frog without chewing; a butterfly draws up nectar through a coiled tube. Despite looking totally different, both events belong to the SAME step of nutrition. Which step, and why are they the same?

- (A) Ingestion — both are simply taking food into the body, whatever its form
- (B) Digestion — both break the food down as it enters
- (C) Absorption — both move nutrients straight into the blood at the mouth
- (D) Egestion — both expel material from the body

5. A textbook lists nutrition steps as: ingestion, digestion, absorption, assimilation, egestion. A student rewrites them as 'ingestion, digestion, assimilation, absorption, egestion.' What is the one error, and why does it matter?

- (A) Egestion is in the wrong place; it should come before digestion
- (B) Ingestion should come last because food is taken in only after digestion
- (C) There is no error; the two orders are equivalent
- (D) Assimilation is placed before absorption, which is impossible because nutrients must enter the blood before cells can use them

6. An Amoeba and a human both complete all five steps of nutrition, yet a biologist says one of them does NOT have a separate organ for each step. Which is correct, and what does the Amoeba use instead?

- (A) The human lacks separate organs and does everything in the stomach
- (B) The Amoeba uses a tiny stomach and intestine inside its cell
- (C) Both have separate organs, so neither statement applies
- (D) The Amoeba lacks separate organs and carries out all five steps within its single cell using temporary structures and a food vacuole

7. A breakfast of bread (mainly starch), egg (mainly protein) and butter (mainly fat) is eaten. In which order are the chemical breakdowns of these three FIRST begun as the meal travels down the gut?

- (A) Protein first (in the mouth), then fat (in the stomach), then starch (in the small intestine)
- (B) Fat first (in the mouth), then starch (in the stomach), then protein (in the small intestine)
- (C) All three begin together in the stomach
- (D) Starch first (in the mouth), then protein (in the stomach), then fat (in the small intestine)

- 8.** A scientist counts an adult human's teeth: she finds 8 teeth used for cutting, 4 for tearing, and the rest for grinding. If the total is 32, how many grinding teeth are there, and what types are they?
- (A) 12 grinding teeth, all molars
 - (B) 16 grinding teeth, made up of canines and incisors
 - (C) 20 grinding teeth, made up of premolars and molars
 - (D) 8 grinding teeth, all premolars
- 9.** A dentist explains that a particular cavity 'has just reached the layer beneath the enamel but has not yet touched the nerves.' Going from the outside inward, which layer has the decay reached, and which is still safe?
- (A) It has reached the dentine; the pulp cavity with the nerves is still safe
 - (B) It has reached the pulp; the dentine is still safe
 - (C) It has reached the enamel only; the dentine is still safe
 - (D) It has reached the root; the crown is still safe
- 10.** A student chews a plain cracker for two full minutes and notices it grows sweeter, then swallows it. They claim 'the sweetness proves protein was digested in my mouth.' What is the correct correction?
- (A) The sweetness proves fat was digested by saliva
 - (B) The sweetness proves protein was indeed digested by saliva
 - (C) Saliva's amylase broke the cracker's starch into sugar; protein digestion does not begin in the mouth
 - (D) The tongue added sugar to the cracker as it was chewed
- 11.** An astronaut drinks water while floating upside-down, with no gravity to pull it down, yet the water still reaches the stomach. The best explanation involves which process, and what does it tell us about that process?
- (A) Gravity — water always falls toward the stomach regardless of body position
 - (B) Peristalsis — rhythmic muscle contractions of the food pipe push food forward independently of gravity
 - (C) Absorption — the food pipe absorbs the water directly into the blood
 - (D) Emulsification — bile in the food pipe pulls the water down
- 12.** Gastric juice contains hydrochloric acid, mucus and pepsin. A student argues the acid's only job is to digest food. Give the most complete correction of the acid's roles.
- (A) The acid directly digests fats into droplets and also cools the food
 - (B) The acid absorbs sugars into the blood and kills germs
 - (C) The acid's only job really is to digest food, so the student is correct
 - (D) The acid kills germs in food AND creates the acidic conditions pepsin needs to work; it does not by itself digest food

13. The stomach lining is itself built largely of protein, and the stomach is full of protein-digesting pepsin and acid. Why does the stomach normally avoid digesting its own wall?

- (A) A layer of mucus coats the inner wall and shields it from the acid and pepsin
- (B) The stomach wall contains no protein, so pepsin has nothing to act on
- (C) Bile from the liver coats the stomach and neutralises the acid
- (D) The wall is made of bone, which pepsin cannot digest

14. A patient produces far too little stomach acid. Of the following, which problem is the MOST directly expected, based on what the acid does?

- (A) Starch can no longer be digested anywhere in the body
- (B) Pepsin works poorly, so protein digestion in the stomach is impaired and food germs are less well killed
- (C) Bile can no longer break fat into droplets
- (D) Water can no longer be absorbed in the large intestine

15. A label is needed for the thick, partly digested, semi-liquid paste that the stomach pushes on into the small intestine. A second label is needed for the partly digested grass a cow brings back to its mouth. The two correct labels are:

- (A) Cud for the stomach's paste; chyme for the cow's grass
- (B) Bile for the stomach's paste; chyme for the cow's grass
- (C) Chyme for the stomach's paste; cud for the cow's regurgitated grass
- (D) Chyme for both, since they are the same material

16. The small intestine of an adult is about 6 metres long and tightly coiled, and its inner wall carries millions of villi. A student says 'all this length and folding is just to store food longer.' What is the better single explanation?

- (A) The length is only to store undigested waste before egestion
- (B) The folding makes the intestine acidic like the stomach
- (C) The great length plus villi hugely increase the surface area and time available for digestion and absorption
- (D) The coiling is needed mainly to churn food the way the stomach does

17. A rare illness causes the villi of a person's small intestine to flatten so the inner surface becomes nearly smooth, although the person keeps eating well. What is the most direct consequence?

- (A) Faster starch digestion in the mouth
- (B) Stronger churning of food by the stomach
- (C) Much less surface for absorption, so the person stays undernourished despite eating
- (D) Increased production of bile by the liver

18. Bile is famous for helping fat digestion, yet bile contains NO digestive enzyme of its own. How can it help, and where does it come from and where does it act?

- (A) Bile, made by the pancreas, chemically breaks fat into amino acids in the stomach
- (B) Bile, made by the liver and acting in the small intestine, breaks large fat globules into tiny droplets so fat-digesting enzymes can work on them more easily
- (C) Bile, made by the gall bladder, turns fat into starch in the mouth
- (D) Bile, made by the liver, absorbs the fat directly into the blood in the stomach

19. After a person's gall bladder is removed, fatty meals trouble them the most, while starchy and protein meals are handled almost normally. What does this pattern reveal about the gall bladder and bile?

- (A) The gall bladder made pepsin, which mainly digests fats
- (B) The gall bladder absorbed starch, so its loss harms starch meals
- (C) Bile mainly digests proteins, which is why protein meals suffer
- (D) The gall bladder stored bile, and bile mainly assists the digestion of fats

20. The pancreas releases a juice into the first part of the small intestine. Compared with most other digestive juices, what is unusual about pancreatic juice?

- (A) It contains no enzymes and only neutralises acid
- (B) It carries enzymes that act on carbohydrates, proteins AND fats, whereas many other juices act on only one or two food types
- (C) It is made and released in the mouth, unlike all other juices
- (D) It acts only on water, leaving food untouched

21. Two organs empty their secretions into the first part of the small intestine to finish digestion. One supplies a fat-emulsifying liquid, the other an enzyme-rich juice for all food types. Name them in that order.

- (A) The liver (bile) first, then the pancreas (pancreatic juice)
- (B) The pancreas (bile) first, then the liver (pancreatic juice)
- (C) The stomach (bile) first, then the salivary glands (pancreatic juice)
- (D) The gall bladder (enzymes) first, then the liver (acid)

22. If the large intestine reabsorbs too LITTLE water from the undigested residue, what is the likely outcome, and how does this contrast with reabsorbing too much?

- (A) Too little water gives loose, watery stools; too much water gives hard, dry stools (constipation)
- (B) Too little water gives hard, dry stools; too much gives watery stools
- (C) Too little water causes vomiting of acid; too much causes faster fat digestion
- (D) Either way the stool is unchanged, since water is not handled here

23. Inside each villus run tiny blood vessels. A meal of rice is digested to glucose, which then crosses the villus wall into these vessels. What step is happening right at the villus, and what comes next for that glucose?

- (A) Digestion is happening at the villus; next the glucose is egested
- (B) Absorption is happening at the villus; next the glucose is carried in the blood to cells, where assimilation occurs
- (C) Egestion is happening at the villus; next the glucose is ingested
- (D) Assimilation is happening at the villus; next the glucose is digested

24. A cow and a human are each fed only fresh grass for a month. The cow thrives; the human grows weak and gets almost no nourishment. What is the best single reason for the difference?

- (A) The human lacks both the rumen microbes and the cud re-chewing system that let a cow break down the cellulose in grass
- (B) The human has no stomach in which to hold the grass
- (C) Grass needs no digestion, so the human simply refuses to eat it
- (D) The human digests the grass in the mouth but the cow does not

25. A ruminant's stomach has four chambers. Microbes first attack the cellulose in the largest chamber, and acid-plus-enzyme digestion (like a human stomach) happens in a different chamber. Name these two chambers in that order.

- (A) Microbial breakdown in the abomasum; acid-and-enzyme digestion in the rumen
- (B) Microbial breakdown in the rumen; acid-and-enzyme digestion in the reticulum
- (C) Microbial breakdown in the rumen; acid-and-enzyme digestion in the abomasum
- (D) Microbial breakdown in the omasum; acid-and-enzyme digestion in the rumen

26. A cow lies resting and slowly chews material it brought back up from its stomach. Why is this re-chewing of the cud useful for getting energy out of grass?

- (A) It turns the cellulose directly into protein as the cow chews
- (B) It makes the grass sweeter so the cow can absorb sugar in its mouth
- (C) It removes the need for the cow to have a stomach at all
- (D) Finer grinding exposes more surface so microbes and enzymes can break the cellulose down more completely

27. An Amoeba surrounds a food particle with its pseudopodia, traps it inside a structure, digests it there, absorbs the useful part, and later throws out the rest. The trapping structure and the final 'throwing out' step are, respectively:

- (A) A food vacuole; and egestion when the vacuole reaches the cell surface
- (B) A gall bladder; and excretion of cell-made waste
- (C) A villus; and absorption of the leftover
- (D) A rumen; and assimilation of the leftover

28. Compare where digestive enzymes act in an Amoeba's food vacuole with where they act in a human. The closest match is that, in a human, enzymes likewise act:

- (A) Inside the cavity of the stomach and intestine, not in the blood
- (B) Inside the blood, after nutrients are absorbed

- (C) On the skin surface, before food is swallowed
- (D) Only inside the brain cells

29. A pseudopodium is a temporary structure, formed only when needed, unlike a human's permanent stomach. What key difference in body plan does this highlight?

- (A) The Amoeba has a permanent organ for every step, just smaller
- (B) The Amoeba digests its food entirely outside its body
- (C) The Amoeba cannot move and so cannot capture food
- (D) The Amoeba makes structures as needed rather than having fixed organs for each step

30. Glucose absorbed from food is finally 'burnt' inside the body's cells to release energy. The structures where this energy release mainly happens, and the step of nutrition this energy use represents, are:

- (A) Mitochondria; and this falls under assimilation, the use of absorbed nutrients
- (B) Villi; and this falls under absorption
- (C) Food vacuoles; and this falls under digestion
- (D) Salivary glands; and this falls under ingestion

31. A six-year-old loses a front tooth and a new one grows in its place; a baby has only 20 teeth while its parent has 32. A student asks why teeth are 'replaced'. The best explanation is:

- (A) Teeth are replaced continuously throughout life, a new one growing every year
- (B) The baby's 20 teeth are permanent and will never change
- (C) Adults grow extra teeth from the food they chew, which is why they have more than babies
- (D) The first set are milk teeth that fall out around age 6–8 and are replaced by a permanent set of up to 32; humans get only these two sets

32. A student claims, 'Because the food pipe just carries food, no muscle is involved — food falls down by gravity.' Two facts disprove this. Which option states them correctly?

- (A) The food pipe is rigid like a bone, so food slides down without muscle
- (B) The food pipe absorbs food into blood, so it never needs to move it down
- (C) The food pipe is filled with bile that washes food into the stomach
- (D) The food pipe has muscular walls that contract in waves (peristalsis), so food moves even against gravity

33. Tooth decay is described as: bacteria feed on sugar left on teeth and release acid that eats into the enamel. Based only on this chain, which action would most directly prevent decay?

- (A) Eating more sugar so the bacteria become full and stop
- (B) Removing the sugar by brushing before the bacteria can turn it into acid

- (C) Brushing to replace the lost enamel with new enamel
- (D) Avoiding all chewing so the enamel is never used

34. A carnivore like a dog has long, pointed teeth beside its front teeth for tearing flesh, while a grass-eater like a cow has broad, flat-topped teeth. The best explanation for this difference is:

- (A) Larger animals always have flatter teeth, regardless of diet
- (B) The kind of food an animal eats shapes the form of its teeth — pointed for tearing flesh, flat for grinding plants
- (C) All mammals have identical teeth; the dog's only look different
- (D) Teeth shape is random and unrelated to what an animal eats

35. Starch digestion begins in the mouth, yet a student is puzzled that it does not finish there, since food is swallowed long before it has all turned to sugar. In which region is starch digestion finally COMPLETED, and which juices act on it there?

- (A) In the stomach, completed by hydrochloric acid and pepsin
- (B) In the small intestine, completed by the juices of the pancreas and the intestine
- (C) In the mouth itself, completed by saliva just before swallowing
- (D) In the large intestine, completed as water is absorbed

36. Three secretions are matched to their food target: saliva, gastric juice, and bile. Which matching is fully correct?

- (A) Saliva starts protein; gastric juice starts fat; bile starts starch
- (B) Saliva starts fat; gastric juice starts starch; bile starts protein
- (C) Saliva, gastric juice, and bile all act on starch only
- (D) Saliva starts starch; gastric juice (pepsin) starts protein; bile emulsifies fat

37. A biologist says, 'In humans the place where food is digested is kept separate from the cells that finally use the nutrients.' Which pairing of locations supports this statement?

- (A) Digestion occurs inside the body cells; nutrients are used in the gut cavity
- (B) Digestion occurs in the cavity of the gut; the nutrients are later used inside body cells reached through the blood
- (C) Digestion occurs in the blood; nutrients are used in the gut cavity
- (D) Digestion and use both occur only in the mouth

38. A patient's small intestine is healthy, but a blockage stops bile from reaching it. Among starch, protein and fat, which food's digestion is most directly hampered, and why?

- (A) Fat, because bile is needed to break fat into small droplets for enzymes to act on
- (B) Starch, because bile is the main starch-digesting enzyme
- (C) Protein, because bile activates pepsin in the small intestine
- (D) All three equally, since bile digests every food type

- 39.** An Amoeba's way of capturing food is compared to four everyday images. Which image best matches engulfing a particle with pseudopodia, and which is clearly wrong?
- (A) Best: a straw sucking up juice; clearly wrong: a hand wrapping a marble
 - (B) Best: a hand cupping and wrapping around a marble; clearly wrong: a straw sucking up juice
 - (C) Best: a knife slicing bread; clearly wrong: a hand wrapping a marble
 - (D) Best: a sieve straining water; clearly wrong: a knife slicing bread
- 40.** A meal moves through three organs in sequence. Which order is correct, given where churning, then most absorption, then water recovery happen?
- (A) Small intestine → stomach → large intestine
 - (B) Large intestine → small intestine → stomach
 - (C) Stomach (churning) → small intestine (most absorption) → large intestine (water recovery)
 - (D) Stomach → large intestine → small intestine
- 41.** A student writes: 'Egestion and assimilation are both about handling the food after digestion.' Which statement best clarifies how they actually differ?
- (A) Assimilation removes undigested leftovers; egestion is cells using nutrients
 - (B) Both are the cells using nutrients, differing only in name
 - (C) Assimilation is cells USING absorbed nutrients; egestion is REMOVING the undigested leftovers
 - (D) Both are the removal of undigested leftovers, differing only in name
- 42.** In an experiment, two test tubes hold starch solution. Tube 1 gets a few drops of saliva and is kept warm; Tube 2 gets only water and is kept warm. After 30 minutes, only Tube 1 tests positive for sugar. What does this best show?
- (A) Warmth alone turns starch into sugar
 - (B) Water breaks starch into sugar over time
 - (C) Starch naturally becomes sugar if simply left to stand
 - (D) Saliva contains an enzyme (amylase) that breaks starch into sugar
- 43.** A scientist measures the inner surface area available for absorption in two model guts of equal length: Gut A has a smooth lining, Gut B has a lining densely covered with villi. Gut B absorbs far more nutrients. The reason is that villi:
- (A) Make the gut more acidic, speeding digestion
 - (B) Shorten the path food must travel, saving time
 - (C) Greatly increase the inner surface area so more nutrients can be absorbed per length
 - (D) Produce extra bile to break down fats
- 44.** A child swallows a small smooth pebble by accident. It cannot be digested. Following the five steps, what most likely happens to it?

- (A) It is digested by stomach acid into nutrients and absorbed
- (B) It is assimilated into the child's growing bones
- (C) It is ingested and finally egested, since the body cannot digest or absorb it
- (D) It is excreted as a cell-made waste product

45. A nutrition diagram labels: liver as the largest gland making a juice, a small sac storing that juice between meals, and a gland whose juice acts on all three food types. The three are, in order:

- (A) Pancreas, rumen, liver
- (B) Liver, pancreas, gall bladder
- (C) Liver, gall bladder, pancreas
- (D) Gall bladder, liver, pancreas

46. A goat regurgitates grass, chews it again, and re-swallows it. A student insists this means goats digest grass entirely in the mouth. What is the correct view?

- (A) Yes, goats fully digest grass in the mouth, so the student is right
- (B) Re-chewing only grinds the cud finer; the real cellulose breakdown depends on microbes in the rumen, not the mouth
- (C) Goats digest grass only in the food pipe as it is re-swallowed
- (D) Goats do not digest grass at all; they merely store it

47. An Amoeba is single-celled, yet it is said to complete the same five steps of nutrition a human does. Which option best captures both the similarity and the difference?

- (A) Both complete all five steps, but the Amoeba does them within one cell while a human uses many specialised organs
- (B) Only the human carries out digestion; the Amoeba never digests
- (C) The Amoeba absorbs food but never egests its leftovers
- (D) The human ingests food but never assimilates it

48. Pepsin in the stomach works best in strongly acidic surroundings, while the enzymes of the small intestine do not. This tells us that for protein digestion to continue smoothly into the small intestine, the chyme should be:

- (A) Made even more acidic, since all digestive enzymes prefer strong acid
- (B) Kept exactly as acidic as the stomach forever
- (C) Dried out completely before entering the small intestine
- (D) Made less acidic in the small intestine, since the intestinal enzymes do not need strong acid

49. A person eats a large fatty meal. Bile is released to help. Tracing the fat: bile breaks it into droplets, enzymes then act, and the products are absorbed. In which organ do all three of these events mainly occur?

- (A) The small intestine
- (B) The stomach
- (C) The mouth
- (D) The large intestine

50. A cow's four-chambered stomach is contrasted with a human's single stomach. The deepest reason a cow needs the extra chambers and microbes is that its food, grass, is rich

in:

- (A) Cellulose, which neither cows nor humans can digest with their own enzymes
- (B) Protein, which only cows can digest
- (C) Sugar, which dissolves instantly in the stomach
- (D) Fat, which the rumen breaks into droplets

51. Two students argue about absorption. One says 'absorption is when cells use the nutrients,' the other says 'absorption is when nutrients pass into the blood.' Who is right, and what is the other actually describing?

- (A) The first student is right; the second is describing digestion
- (B) The second student is right (absorption is nutrients passing into blood); the first is describing assimilation
- (C) Both are right, since absorption and assimilation mean the same
- (D) Neither is right; absorption is the removal of waste

52. A diagram shows food taking 4 minutes in the mouth-and-food-pipe region, a few hours in a churning bag, then many hours in a long coiled tube where most nutrients leave for the blood. The long coiled tube is the:

- (A) Oesophagus
- (B) Stomach
- (C) Small intestine
- (D) Large intestine

53. An Amoeba forms a pseudopodium to capture food. The same kind of extension is also useful to the Amoeba for another life activity. Which is it, and what does this show about pseudopodia?

- (A) Movement — pseudopodia let the Amoeba both move about and capture food
- (B) Producing bile — pseudopodia make bile while also capturing food
- (C) Absorbing oxygen only — pseudopodia are used solely for breathing
- (D) Storing waste forever — pseudopodia hold undigested matter permanently

54. A meal of sweet, watery fruit is eaten. Comparing where its sugar is mainly absorbed with where its water is mainly absorbed, the correct pairing is:

- (A) Sugar mainly absorbed in the large intestine; water in the small intestine
- (B) Both sugar and water mainly absorbed in the stomach
- (C) Sugar mainly absorbed in the small intestine; water mainly absorbed in the large intestine
- (D) Both sugar and water mainly absorbed in the mouth

55. Suppose a medicine completely stops peristalsis in the food pipe but leaves everything else working. What is the most direct effect on a person trying to eat?

- (A) Food would be digested faster in the stomach
- (B) Swallowed food would struggle to move down the food pipe to the stomach
- (C) More water would be absorbed in the large intestine
- (D) Starch digestion in the mouth would stop

- 56.** A person's pancreatic juice fails to reach the small intestine, though saliva and gastric juice are normal. Which broad consequence best fits, given what the pancreatic juice does?
- (A) Only starch digestion is impaired, and only in the mouth
 - (B) Only protein digestion is impaired, and only in the stomach
 - (C) Only water absorption in the large intestine is impaired
 - (D) Digestion of carbohydrates, proteins and fats in the small intestine is all impaired
- 57.** A student lists the human teeth: 'incisors for cutting, canines for tearing, premolars and molars for grinding.' A friend objects that canines are for grinding. Who is right and why?
- (A) The friend is right — canines are broad and flat for grinding
 - (B) Both are wrong — incisors are for grinding, not cutting
 - (C) Both are wrong — molars are for cutting, not grinding
 - (D) The student is right — canines are pointed teeth for tearing, while the flat premolars and molars do the grinding
- 58.** Glucose, amino acids and other simple molecules cross from the gut into the blood at the villi. A student says 'this proves the villi digest the food.' Correct this.
- (A) The villi both digest and then store the food permanently
 - (B) The villi do not digest food; they absorb the already-digested simple molecules into the blood
 - (C) The villi digest food but do not absorb anything
 - (D) The villi only churn food, like the stomach
- 59.** A cow eats grass in the morning, then rests in the afternoon chewing cud. A student concludes the cow eats the same grass twice from outside. What is the accurate description?
- (A) The cow finds and eats fresh grass a second time in the afternoon
 - (B) The cow chews its own saliva, not grass, in the afternoon
 - (C) The cow brings already-swallowed grass back up from its stomach to chew again; it is not fresh grass eaten twice
 - (D) The cow chews grass that grew inside its stomach during the day
- 60.** Putting it together: in a human, name the place where protein digestion **BEGINS**, the place where fat digestion is **COMPLETED**, and the place where water is chiefly **RECOVERED**, in that order.
- (A) Stomach; small intestine; large intestine
 - (B) Mouth; stomach; small intestine
 - (C) Small intestine; mouth; stomach
 - (D) Large intestine; stomach; mouth

Solutions — Science (Class 7), Chapter 2:

Nutrition in Animals — Paper 3

Paper 3 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	11	B	21	A	31	D	41	C	51	B
2	B	12	D	22	A	32	D	42	D	52	C
3	C	13	A	23	B	33	B	43	C	53	A
4	A	14	B	24	A	34	B	44	C	54	C
5	D	15	C	25	C	35	B	45	C	55	B
6	D	16	C	26	D	36	D	46	B	56	D
7	D	17	C	27	A	37	B	47	A	57	D
8	C	18	B	28	A	38	A	48	D	58	B
9	A	19	D	29	D	39	B	49	A	59	C
10	C	20	B	30	A	40	C	50	A	60	A

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (D) Glucose entering the blood is absorption; its later use to build new body tissue (growth) is assimilation, so the correct pair in order is absorption then assimilation. 'Digestion then absorption' is wrong because the blood-appearance step is absorption, not digestion — digestion happened earlier, before glucose existed. 'Absorption then egestion' is wrong because growth uses the nutrient, it does not throw it out as waste. 'Assimilation then absorption' reverses the order — a nutrient must enter the blood (absorption) before cells can use it (assimilation).

2. (B) Glucose reaches the cells (so it was absorbed into blood) but the cells cannot use it — assimilation is the step that fails, while absorption clearly works. 'Absorption is failing' is wrong because glucose does reach the cells, proving it crossed into the blood. 'Digestion is failing' contradicts the statement that food is broken down normally. 'Egestion is failing' is irrelevant — the problem is the cells using nutrients, not removing undigested waste.

3. (C) Egestion (X) throws out undigested material that never entered cells, whereas excretion (Y) removes wastes such as carbon dioxide produced by cells — so X is egestion and Y is excretion. 'Both are egestion' is wrong because cell-made waste is excretion, a different process. The two 'swapped' options are wrong because they reverse which process handles undigested fibre versus cell-made waste.

4. (A) Ingestion just means taking food into the body, and both swallowing a frog and sipping nectar do exactly that, so it is ingestion. 'Digestion' is wrong because neither act is the chemical breakdown of food; that comes after. 'Absorption' is wrong because the mouth does not pass these foods directly into blood. 'Egestion' is wrong because food is going in, not out.

5. (D) The student swapped absorption and assimilation; since cells can only use nutrients after those nutrients have entered the blood, absorption must precede assimilation. 'Egestion before digestion' is wrong because you cannot remove undigested residue before food is even broken down. 'Ingestion last' is absurd — food must be taken in first. 'No error' is wrong because the swapped order breaks the cause-and-effect chain.

6. (D) An Amoeba is one cell, so it performs every step inside that cell using a pseudopodium and a food vacuole rather than dedicated organs — that is the correct picture. 'The human lacks organs' is backwards; humans have many organs. 'A tiny stomach and intestine' is wrong because a single cell has no such organs. 'Both have separate organs' is wrong because the Amoeba clearly does not.

7. (D) Saliva starts starch digestion in the mouth, stomach acid and pepsin start protein digestion, and fat breakdown chiefly begins in the small intestine with bile and pancreatic juice — so the order is starch, protein, fat. The other options misassign the starting organs: protein and fat do not begin in the mouth (only starch does), and the three do not all begin together in the stomach, since starch already started in the mouth and major fat digestion waits for the small intestine.

8. (C) Cutting teeth are incisors (8) and tearing teeth are canines (4), leaving $32 - 8 - 4 = 20$ grinding teeth, which are the premolars and molars. '12, all molars' ignores the 8 premolars that also grind. '16, canines and incisors' is wrong because those are cutting/tearing teeth, not grinding. '8, all premolars' leaves out the 12 molars.

9. (A) Below the hard enamel lies the dentine, and deeper still is the pulp cavity holding the nerves; decay 'beneath the enamel but not at the nerves' is therefore in the dentine with the pulp still safe. 'Reached the pulp' is wrong because the nerves are untouched. 'Enamel only' contradicts 'beneath the enamel.' 'Root vs crown' confuses position in the jaw with the inward layering of a tooth.

10. (C) Crackers are starchy, and salivary amylase converts starch into simpler sugars, which tastes sweet — protein digestion starts later, in the stomach. 'Fat was digested' is

wrong; saliva does not act on fat. 'Protein was digested' repeats the student's error — the enzyme in saliva works on starch, not protein. 'The tongue added sugar' is wrong; the sugar comes from breaking down the cracker's own starch.

11. (B) The wave-like muscular contractions called peristalsis push food along the food pipe even without gravity's help, which is why an upside-down astronaut can still swallow. 'Gravity' is contradicted by the upside-down, weightless setting. 'Absorption' is wrong; the food pipe does not absorb water into blood. 'Emulsification' involves bile acting on fat in the small intestine, not water in the food pipe.

12. (D) Stomach acid does two things — it kills many germs in the food and it makes the stomach acidic so pepsin can act on protein; the acid itself is not the digesting enzyme. 'Digests fats and cools food' is wrong; fat-droplet formation is bile's job and acid does not cool food. 'Absorbs sugars' is wrong; the stomach is not where sugars are absorbed. 'Only digests food' is the misconception being corrected.

13. (A) Mucus secreted by the stomach lining forms a protective barrier between the wall and the acidic, pepsin-rich juice, preventing self-digestion. 'No protein' contradicts the stem, which says the wall is mostly protein. 'Bile coats the stomach' is wrong; bile is made in the liver and acts in the small intestine. 'Made of bone' is wrong; the stomach wall is muscle and soft tissue, not bone.

14. (B) Acid both activates the right conditions for pepsin and kills germs, so too little acid most directly weakens protein digestion in the stomach and germ-killing. 'Starch cannot be digested' is wrong; starch digestion uses amylase in the mouth and small intestine, not stomach acid. 'Bile cannot break fat' is wrong; bile's action does not depend on stomach acid. 'Water cannot be absorbed' is unrelated to stomach acid.

15. (C) The stomach's semi-liquid output is called chyme, while the grass a ruminant brings back up to chew is called cud — two different things. Swapping the names (cud for the paste, chyme for the grass) is the classic mix-up and is wrong. 'Bile' is neither; bile is a liver secretion. 'Chyme for both' is wrong because cud is regurgitated grass in a ruminant, not the human stomach's paste.

16. (C) Length and villi together maximise the surface for absorbing nutrients and give enough time to finish digestion, which is the real purpose. 'Store waste' is wrong; storing undigested waste is the large intestine's role, not the small intestine's. 'Make it acidic' is wrong; the small intestine is not acidic. 'To churn' is wrong; churning is the stomach's job.

17. (C) Villi provide the surface for absorbing nutrients, so flattening them cuts absorption sharply and the person grows undernourished even while eating. 'Faster mouth digestion' is unrelated to the intestinal wall. 'Stronger stomach churning' is unrelated; the stomach is upstream and unaffected. 'More bile' has nothing to do with villi being flattened.

18. (B) Bile is produced by the liver, acts in the small intestine, and physically breaks fat into small droplets to expand the surface for enzymes — it needs no enzyme of its own. 'Made by the pancreas, into amino acids in the stomach' is wrong on the source, product (amino acids come from protein), and place. 'Turns fat into starch' is chemically impossible. 'Absorbs fat in the stomach' is wrong; bile does not absorb, and this happens in the small intestine.

19. (D) Because only fatty meals become hard to handle, bile (stored in the gall bladder) must mainly aid fat digestion. 'Made pepsin' is wrong; pepsin is a stomach enzyme for protein, not made by the gall bladder. 'Absorbed starch' is wrong and would harm starch meals, which are fine here. 'Bile digests proteins' contradicts the observation that protein meals are normal.

20. (B) Pancreatic juice is notable for containing enzymes for all three main food types at once, while juices like saliva (starch) or gastric juice (protein) are more limited. 'No enzymes' contradicts its enzyme-rich nature. 'Made in the mouth' is wrong; it is made by the pancreas and acts in the small intestine. 'Acts only on water' is wrong; it acts on carbohydrates, proteins and fats.

21. (A) Bile (the fat-emulsifying liquid) is made by the liver, and the enzyme-rich juice for all food types is from the pancreas. Swapping their products (pancreas-makes-bile, liver-makes-pancreatic-juice) is wrong. The stomach and salivary glands do not empty into the small intestine. The gall bladder only stores bile, not enzymes, and the liver does not make acid.

22. (A) The large intestine's job is to absorb water; absorbing too little leaves the residue watery (loose stools), and absorbing too much leaves it hard and dry (constipation). Reversing these effects is wrong. Vomiting acid and fat digestion are unrelated to large-intestine water handling. 'Unchanged' is wrong because water absorption is exactly what this organ does.

23. (B) Glucose passing through the villus into blood is absorption, and the cells then use it (assimilation) after it is carried there. 'Digestion at the villus, then egested' is wrong on both counts; glucose is already digested, and useful glucose is not thrown out. 'Egestion then ingestion' reverses everything. 'Assimilation at the villus' is wrong; assimilation happens in the cells, after absorption, and glucose is not digested again.

24. (A) Cows digest grass because microbes in their rumen break down cellulose and they re-chew cud; humans have neither, so they get little from grass. 'No stomach' is false; humans have a stomach. 'Grass needs no digestion' is wrong; its cellulose must be broken down. 'Human digests grass in the mouth' is wrong; humans cannot digest cellulose anywhere.

25. (C) Cellulose is first broken down by microbes in the large rumen, and the human-stomach-like acid-and-enzyme digestion occurs in the abomasum. Swapping rumen and abomasum is the classic error. The reticulum and omasum are the other chambers and are not where the human-like acid digestion mainly occurs.

26. (D) Re-chewing grinds the grass finer, increasing surface area so microbes and enzymes act more effectively on the tough cellulose. 'Turns cellulose into protein' is false; chewing does not transmute one nutrient into another. 'Sweeter, absorbed in the mouth' is wrong; cows do not absorb sugar in the mouth. 'Removes the need for a stomach' is wrong; the stomach chambers are essential.

27. (A) Amoeba digests its meal inside a food vacuole and later expels the undigested residue as egestion when the vacuole reaches the surface. 'Gall bladder' and 'excretion of cell waste' are wrong; Amoeba has no gall bladder and this is undigested food (egestion), not cell-made waste. 'Villus / absorption' is wrong; villi are human structures and the leftover is removed, not absorbed. 'Rumen / assimilation' is wrong; rumens are in ruminants and the leftover is not used by the cell.

28. (A) In both, digestion happens in an enclosed space outside the body's cells — the food vacuole in Amoeba and the cavity of the stomach and intestine in humans. 'Inside the blood' is wrong; the blood carries absorbed nutrients, it does not digest them. 'On the skin surface' is wrong; humans do not digest food on their skin. 'Inside the brain' is unrelated to digestion.

29. (D) Being single-celled, the Amoeba forms structures like pseudopodia on demand instead of keeping fixed organs as humans do. 'Permanent organ for every step' contradicts the temporary nature of the pseudopodium. 'Digests outside its body' is wrong; digestion happens inside the food vacuole. 'Cannot move' is wrong; the pseudopodia also let it move and engulf food.

30. (A) Cells release energy from glucose mainly in their mitochondria, and using absorbed nutrients this way is part of assimilation. 'Villi / absorption' is wrong; villi absorb nutrients, they do not burn them, and absorption is the prior step. 'Food vacuoles / digestion' is wrong; food vacuoles are Amoeba structures, and burning glucose is not digestion. 'Salivary glands / ingestion' is unrelated to releasing energy in cells.

31. (D) Humans have just two sets of teeth in a lifetime: the milk teeth (about 20) appear in early childhood and begin to fall out around age 6–8, and are replaced by a permanent set of up to 32. So the child's lost front tooth is a milk tooth giving way to a permanent one, and the parent's larger count is the completed permanent set. (A) is wrong because teeth are replaced only once, not continuously year on year. (B) is wrong because the baby's teeth are milk teeth and do change. (C) is wrong because teeth do not grow from chewed food; the extra adult teeth come with the permanent set.

32. (D) The food pipe's muscular walls squeeze in waves (peristalsis), moving food along even when gravity cannot help. 'Rigid like a bone' is wrong; it is muscular and flexible. 'Absorbs food into blood' is wrong; the food pipe does not absorb nutrients. 'Filled with bile' is wrong; bile is in the small intestine, not the food pipe.

33. (B) The chain starts with sugar feeding bacteria, so removing the sugar by brushing breaks the chain and prevents acid attack. 'Eating more sugar' would feed the bacteria more, worsening it. 'Replace lost enamel' is wrong; brushing cannot regrow enamel. 'Avoiding chewing' does nothing about the sugar the bacteria feed on.

34. (B) Tooth shape matches diet — sharp pointed teeth tear meat, broad flat teeth grind plant food. 'Larger animals always flatter' is false; shape tracks diet, not size. 'Identical teeth' contradicts the stated difference. 'Random and unrelated' ignores the clear food-to-tooth match.

35. (B) Salivary amylase starts breaking starch into sugar in the mouth, but the food is swallowed within seconds — long before the change is complete — and the stomach's acid then halts the amylase. Starch digestion is finished only in the small intestine, where the pancreatic juice and the intestinal juice act on it. (A) is wrong: the stomach's acid and pepsin act on protein, not starch, and the acid in fact stops amylase. (C) is wrong: the mouth only begins starch digestion; food is swallowed before it can finish there. (D) is wrong: the large intestine absorbs water and does not complete digestion.

36. (D) Saliva's amylase acts on starch, gastric pepsin on protein, and bile emulsifies fat — the correct matching. The two 'rotated' options misassign each secretion to the wrong food type. 'All act on starch' is wrong; only saliva acts on starch, while gastric juice handles protein and bile handles fat.

37. (B) Food is digested in the gut cavity, then absorbed nutrients travel by blood to cells where they are used — keeping the two places separate. 'Digestion inside cells' reverses it; cells use, not digest, the food. 'Digestion in the blood' is wrong; blood transports absorbed nutrients. 'Both in the mouth' is wrong; most digestion and all cell use occur far from the mouth.

38. (A) Bile emulsifies fat into droplets, so blocking bile most directly harms fat digestion. 'Starch' is wrong; bile is not a starch enzyme (amylase does that). 'Protein' is wrong; pepsin acts in the stomach, not via bile. 'All three equally' is wrong; bile is specifically tied to fat handling.

39. (B) Pseudopodia surround and enclose the food, just like a hand cupping a marble; a straw sucking juice is the opposite of engulfing, so it is clearly wrong. The option naming the straw as 'best' reverses the right answer. A knife slicing bread describes cutting, not engulfing, so it is not the best match. A sieve straining water describes filtering, which is also not engulfing.

40. (C) Food is churned in the stomach, then most nutrients are absorbed in the small intestine, and finally water is recovered in the large intestine — that order. The alternatives misplace the stomach after the small intestine, run the sequence backwards, or put the large intestine before the small intestine, none of which match the real path of food.

41. (C) Assimilation is the constructive use of absorbed nutrients by cells, while egestion is throwing out the undigested residue — opposite in purpose. The first distractor swaps the two definitions. The other two wrongly claim the terms mean the same thing, when one is using nutrients and the other is removing waste.

42. (D) Only the tube with saliva produced sugar, so something in saliva — the enzyme amylase — converted starch to sugar. 'Warmth alone' is ruled out because both tubes were warm but only Tube 1 changed. 'Water breaks starch' is ruled out because Tube 2 had water and no change. 'Naturally becomes sugar' is ruled out because the water tube showed no sugar.

43. (C) Villi multiply the absorbing surface, so the villi-lined gut absorbs more even at the same length. 'More acidic' is wrong; villi do not change acidity. 'Shorten the path' is wrong; villi add surface, they do not shorten the gut. 'Produce bile' is wrong; bile comes from the liver, not the villi.

44. (C) An indigestible pebble is taken in (ingestion) and later removed as undigested matter (egestion), with no digestion, absorption or assimilation. 'Digested into nutrients' is wrong; a pebble cannot be broken down by stomach acid into food. 'Assimilated into bones' is wrong; only absorbed nutrients can be assimilated. 'Excreted as cell waste' is wrong; this is undigested matter (egestion), not a waste made inside cells.

45. (C) The liver is the largest gland and makes bile, the gall bladder stores bile between meals, and the pancreas makes a juice acting on all three food types — in that order. 'Pancreas, rumen, liver' is wrong; the rumen is a cow's stomach chamber, not a gland, and the order is muddled. 'Liver, pancreas, gall bladder' wrongly makes the pancreas the storage sac. 'Gall bladder, liver, pancreas' wrongly makes the gall bladder the largest gland.

46. (B) Re-chewing the cud grinds grass finer, but the cellulose is actually broken down by microbes in the rumen, not in the mouth. 'Fully digest in the mouth' is the misconception being corrected. 'Only in the food pipe' is wrong; the food pipe transports, it does not digest cellulose. 'Do not digest at all' is wrong; ruminants do digest grass, thanks to rumen microbes.

47. (A) Both perform ingestion through egestion, but the Amoeba does it all in one cell while a human spreads the work across organs. 'Amoeba never digests' is wrong; it digests inside its food vacuole. 'Never egests' is wrong; it expels undigested matter at the cell surface. 'Human never assimilates' is wrong; humans clearly use nutrients to grow and repair.

48. (D) Stomach pepsin needs acid, but the small intestine's enzymes do not, so the chyme is made less acidic on entering the small intestine. 'Even more acidic' is wrong; not all enzymes like strong acid — the intestinal ones do not. 'Kept exactly as acidic' is wrong; the environment changes between organs. 'Dried out completely' is wrong; digestion needs a watery medium.

49. (A) Bile emulsification, enzyme action on fat, and absorption of the products all mainly happen in the small intestine. 'The stomach' is wrong; bile and the fat-digesting enzymes act in the small intestine, not the stomach. 'The mouth' handles only starch and does not digest or absorb fat. 'The large intestine' chiefly absorbs water, not fat products.

50. (A) Grass is rich in cellulose, which no mammal's own enzymes break down, so cows rely on rumen microbes — hence the extra chambers. 'Protein' is wrong; protein is not the special challenge of grass and is handled by ordinary digestion. 'Sugar dissolving instantly' is wrong; grass is not mainly sugar. 'Fat broken by the rumen' is wrong; the rumen's special job is cellulose breakdown by microbes, not fat.

51. (B) Absorption is nutrients entering the blood, so the second student is correct, and the first student is describing assimilation (cells using nutrients). 'The first is right, second describes digestion' reverses the truth and mislabels absorption as digestion. 'Both are right / same thing' is wrong; absorption and assimilation are distinct steps. 'Removal of waste' is egestion or excretion, not absorption.

52. (C) The long coiled tube where most nutrients are absorbed into the blood is the small intestine. 'Oesophagus' is wrong; it is the short food pipe where food spends only seconds to minutes. 'Stomach' is the churning bag, described separately. 'Large intestine' mainly absorbs water, not most nutrients.

53. (A) Pseudopodia serve a double duty: they let the Amoeba move and also engulf food. 'Producing bile' is wrong; Amoeba has no liver or bile. 'Absorbing oxygen only' is wrong; pseudopodia are not solely breathing organs and they do capture food. 'Storing waste forever' is wrong; undigested matter is egested, not stored permanently.

54. (C) Digested sugar is mainly absorbed in the small intestine, while most remaining water is absorbed later in the large intestine. The reversed option misplaces both. 'Both in the stomach' is wrong; the stomach mainly churns and starts protein digestion. 'Both in the mouth' is wrong; the mouth neither absorbs sugar nor most water.

55. (B) Peristalsis pushes food down the food pipe, so stopping it most directly blocks food from reaching the stomach. 'Digested faster in the stomach' is wrong; stomach digestion is unaffected by the food pipe's muscle waves. 'More water absorbed' in the large intestine is unrelated. 'Starch digestion in the mouth stops' is wrong; salivary amylase still works in the mouth.

56. (D) Pancreatic juice carries enzymes for carbohydrates, proteins and fats, so losing it impairs digestion of all three in the small intestine. 'Only starch in the mouth' is wrong; that is salivary amylase, which is normal here. 'Only protein in the stomach' is wrong; that is gastric pepsin, also normal. 'Only water absorption' is wrong; that is the large intestine, unrelated to pancreatic enzymes.

57. (D) Canines are sharp, pointed teeth used for tearing, while the broad premolars and molars grind, so the student is correct. The friend is wrong because canines are not flat grinders. 'Incisors for grinding' is wrong; incisors cut. 'Molars for cutting' is wrong; molars grind.

58. (B) By the time molecules reach the villi they are already digested; the villi's job is to absorb them into the blood, not to digest. 'Digest and store permanently' is wrong on both counts; villi absorb, they do not store. 'Digest but not absorb' is the reverse of their true role. 'Only churn food' is wrong; churning is the stomach's job, not the villi's.

59. (C) Cud is partly digested grass regurgitated from the stomach and re-chewed, not new grass taken in from outside. 'Fresh grass a second time' is wrong; the cud comes back up from the stomach. 'Chews its own saliva' is wrong; it chews regurgitated grass. 'Grass that grew inside the stomach' is impossible; grass does not grow inside an animal.

60. (A) Protein digestion begins in the stomach, fat digestion is completed in the small intestine, and most water is recovered in the large intestine — that order. 'Mouth; stomach; small intestine' is wrong; protein does not begin in the mouth and water is not chiefly recovered in the small intestine. The remaining options jumble the organs so each role is in the wrong place.